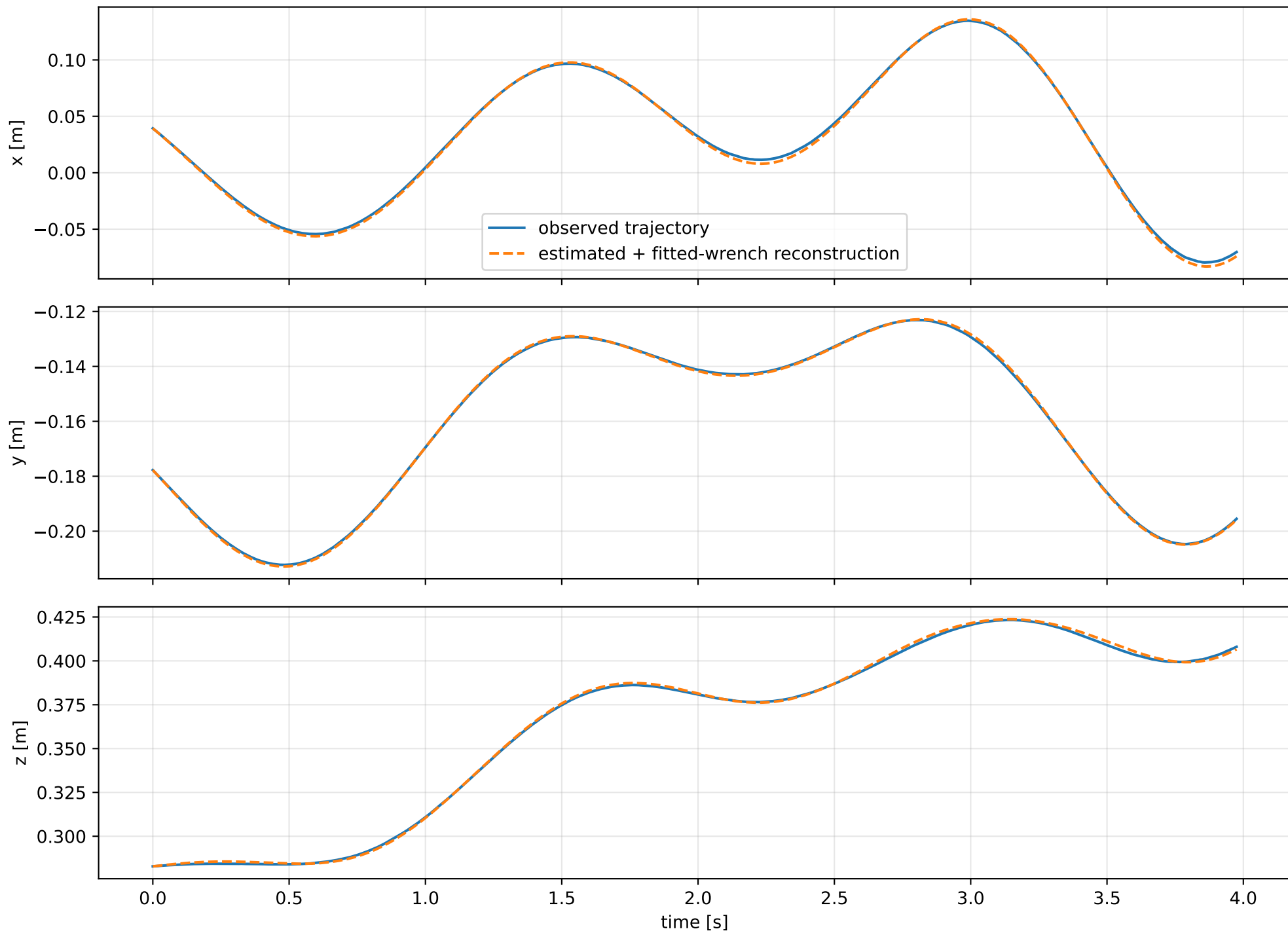


sweep_sg_window_covariance__005: trajectory

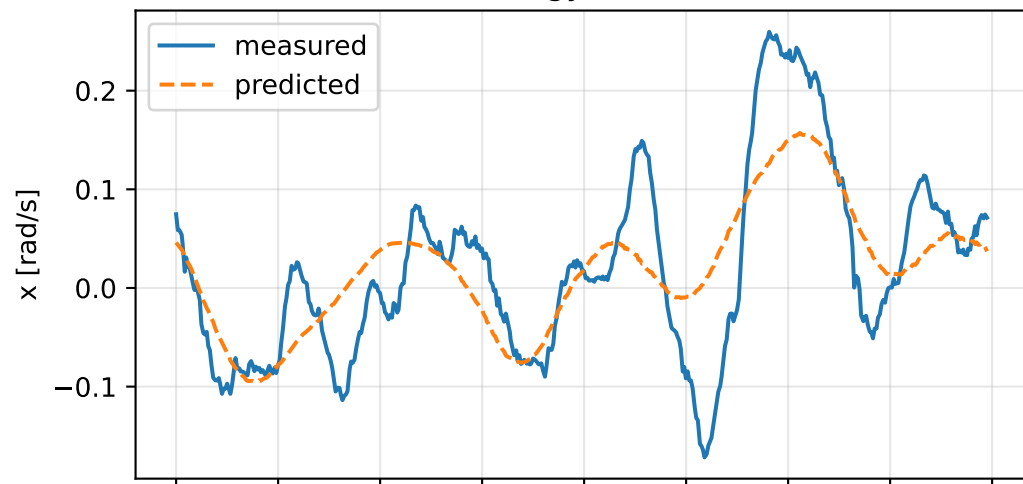


sweep_sg_window_covariance_005: residual wrench

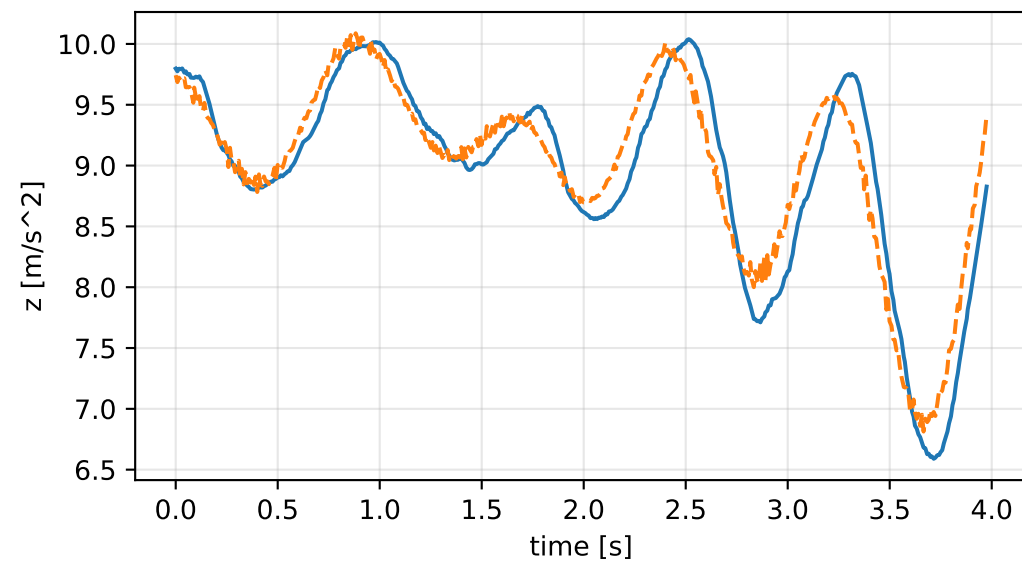
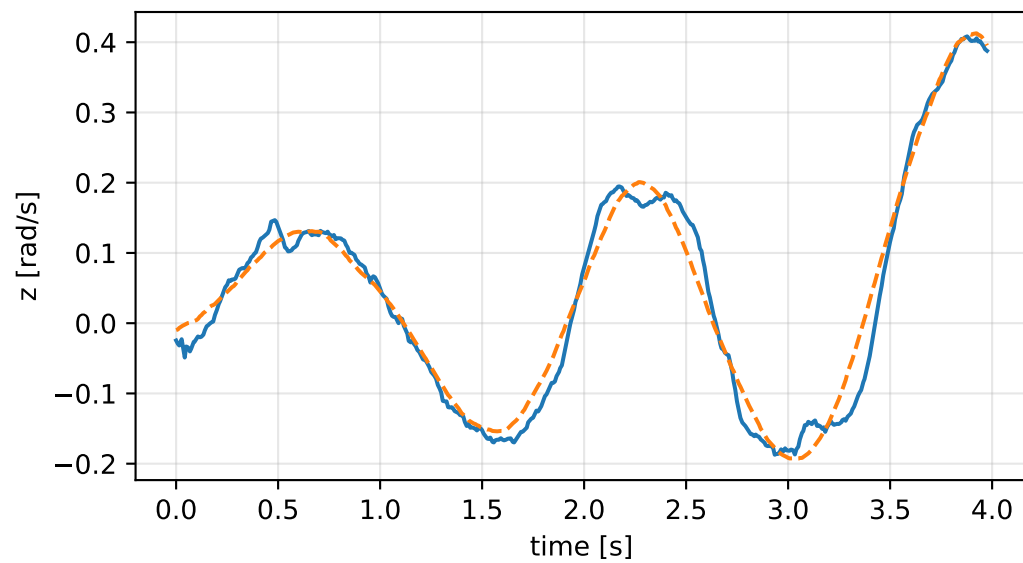
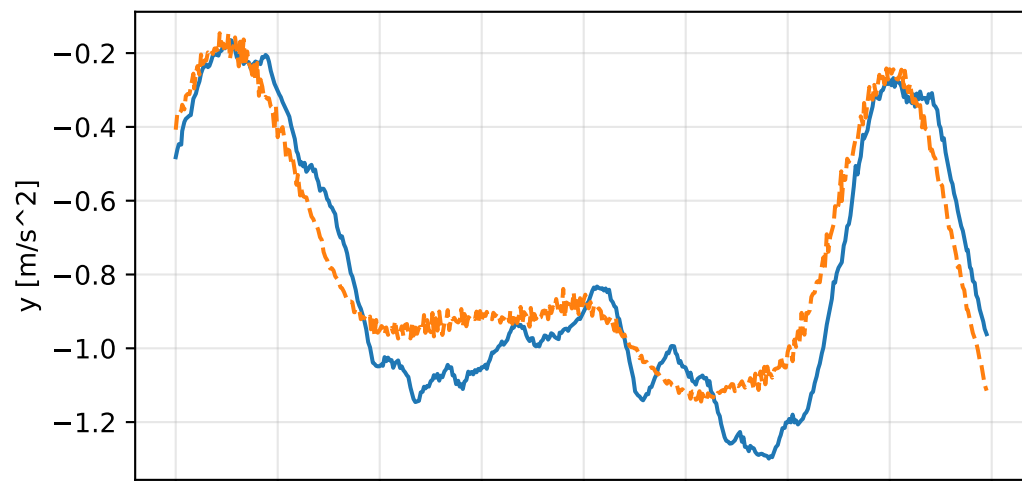
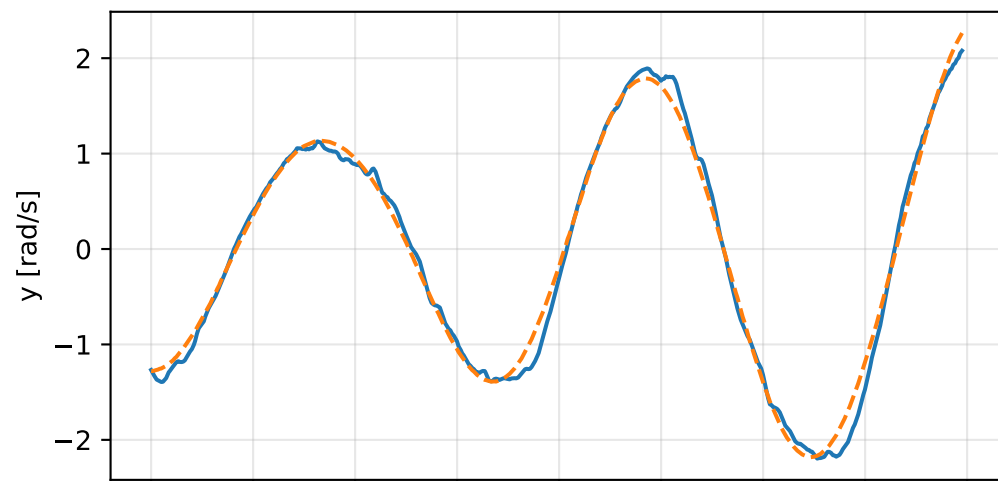
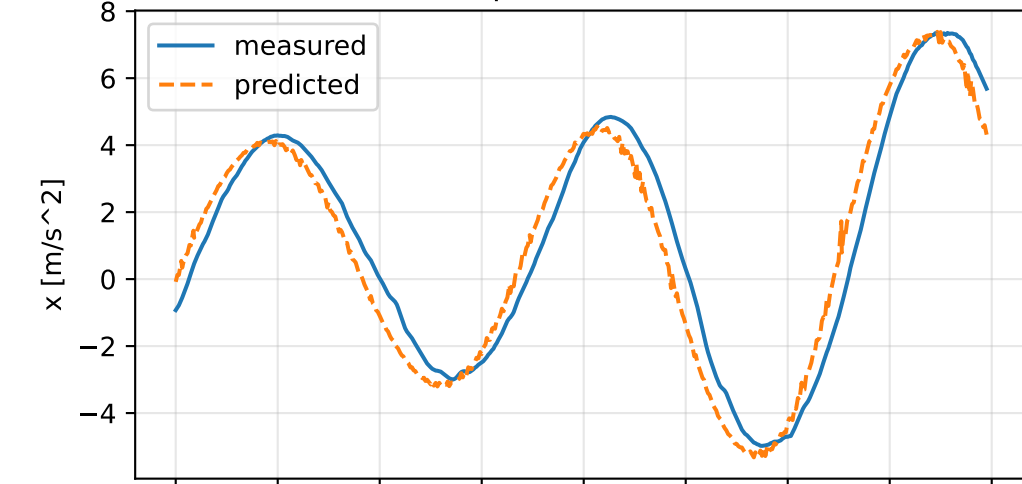


sweep_sg_window_covariance_005: measured sensor comparison

gyro



specific force



sweep_sg_window_covariance__005: nominal -> estimated parameters

Case: sweep_sg_window_covariance__005
status: completed (all stages completed)

mass [kg] nominal 2.35159759 estimated 1.07350525 delta -1.27809234

inertia [kg m^2] (nominal -> estimated; delta)

[6.5000061e-02 -7.2789925e-07 1.9015080e-05] -> [0.2363592 0.0025952 -0.0458541] d=[0.1713592 0.0025959 -0.0458731]
[-7.2789925e-07 6.4952656e-02 5.9167305e-08] -> [0.0025952 0.121431 -0.0317584] d=[0.0025959 0.0564783 -0.0317584]
[1.9015080e-05 5.9167305e-08 1.2899211e-01] -> [-0.0458541 -0.0317584 0.2265343] d=[-0.0458731 -0.0317584 0.0975422]

CoG [m] [-2.0247086e-03 -3.0526579e-05 9.5097496e-03] -> [-0.0219886 0.0426704 -0.0628149] d=[-0.0199639 0.0427009

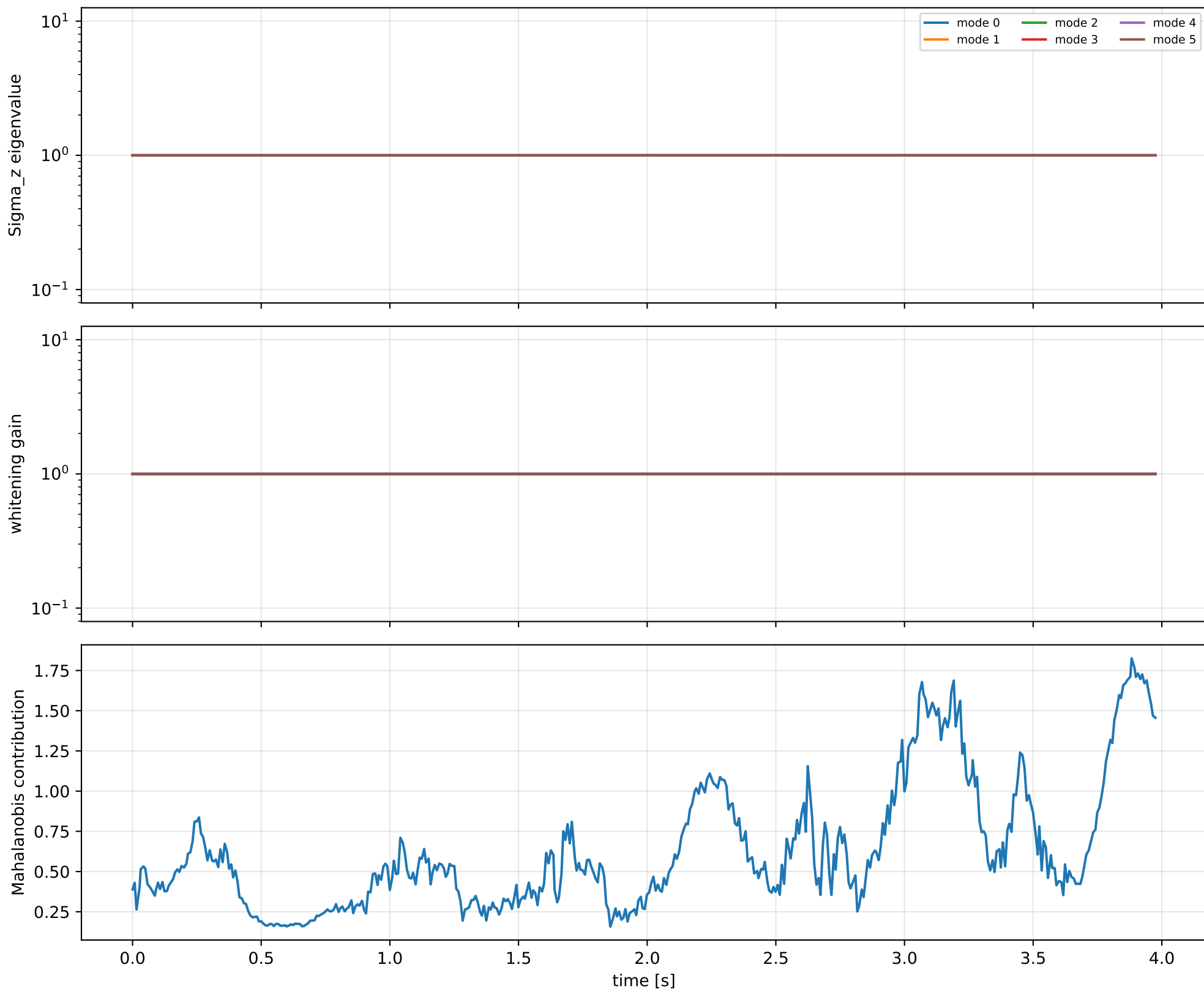
rotor force effectiveness

[1. 1. 1. 1.] -> [0.3624545 0.3140368 0.4128156 0.4611062] d=[-0.6375455 -0.6859632 -0.5871844 -0.5388938]

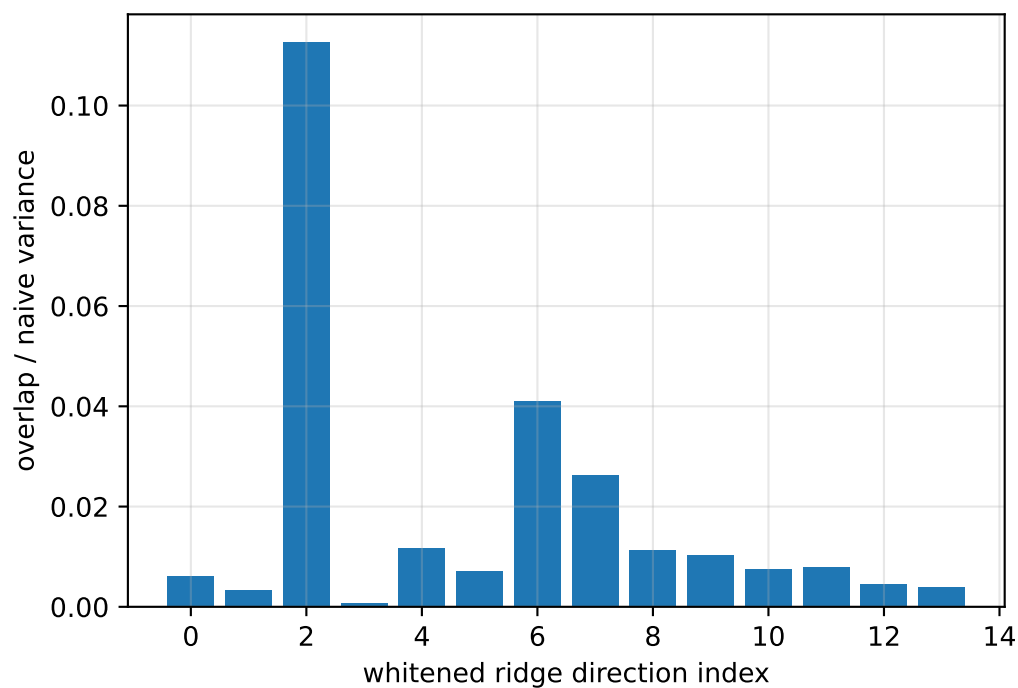
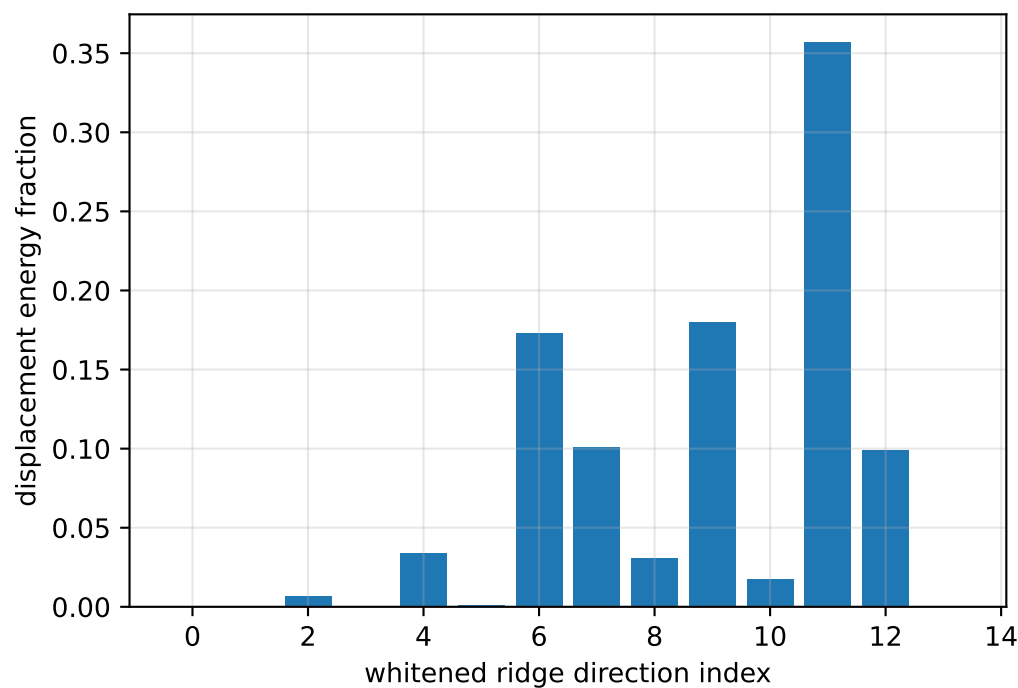
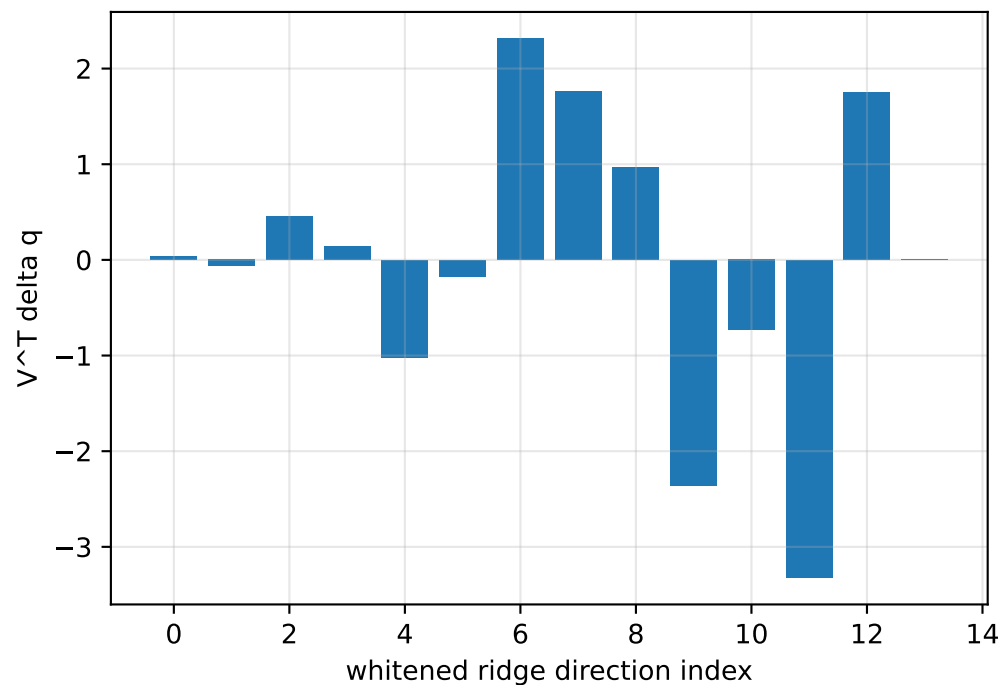
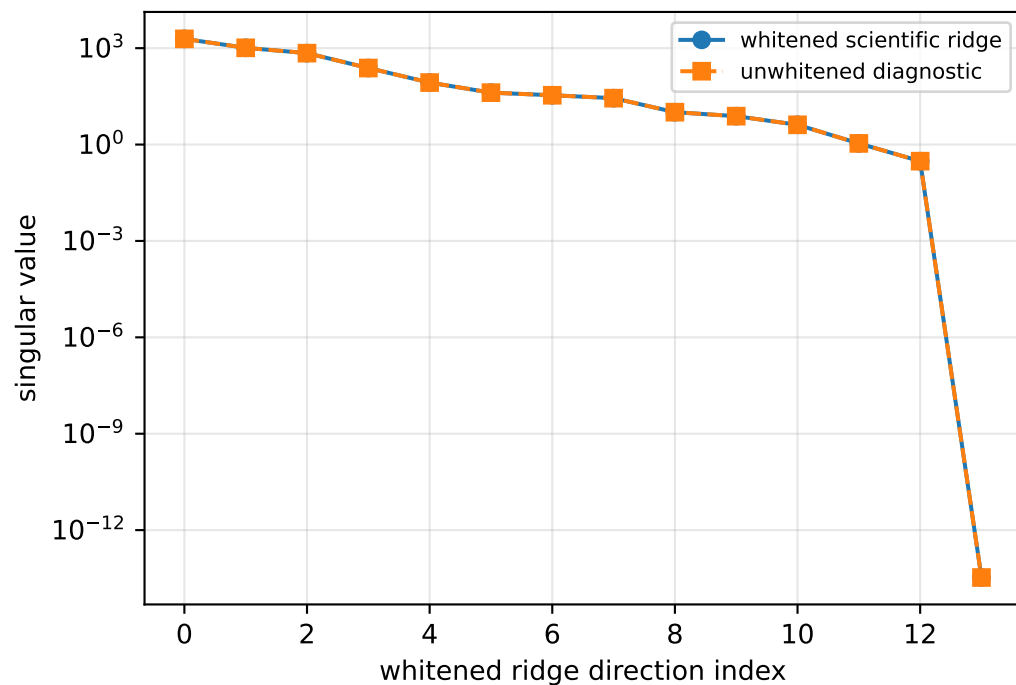
rotor lag [s] 0.185006809

gimbal lag [s] 0.0449211735

sweep_sg_window_covariance__005: covariance weighting

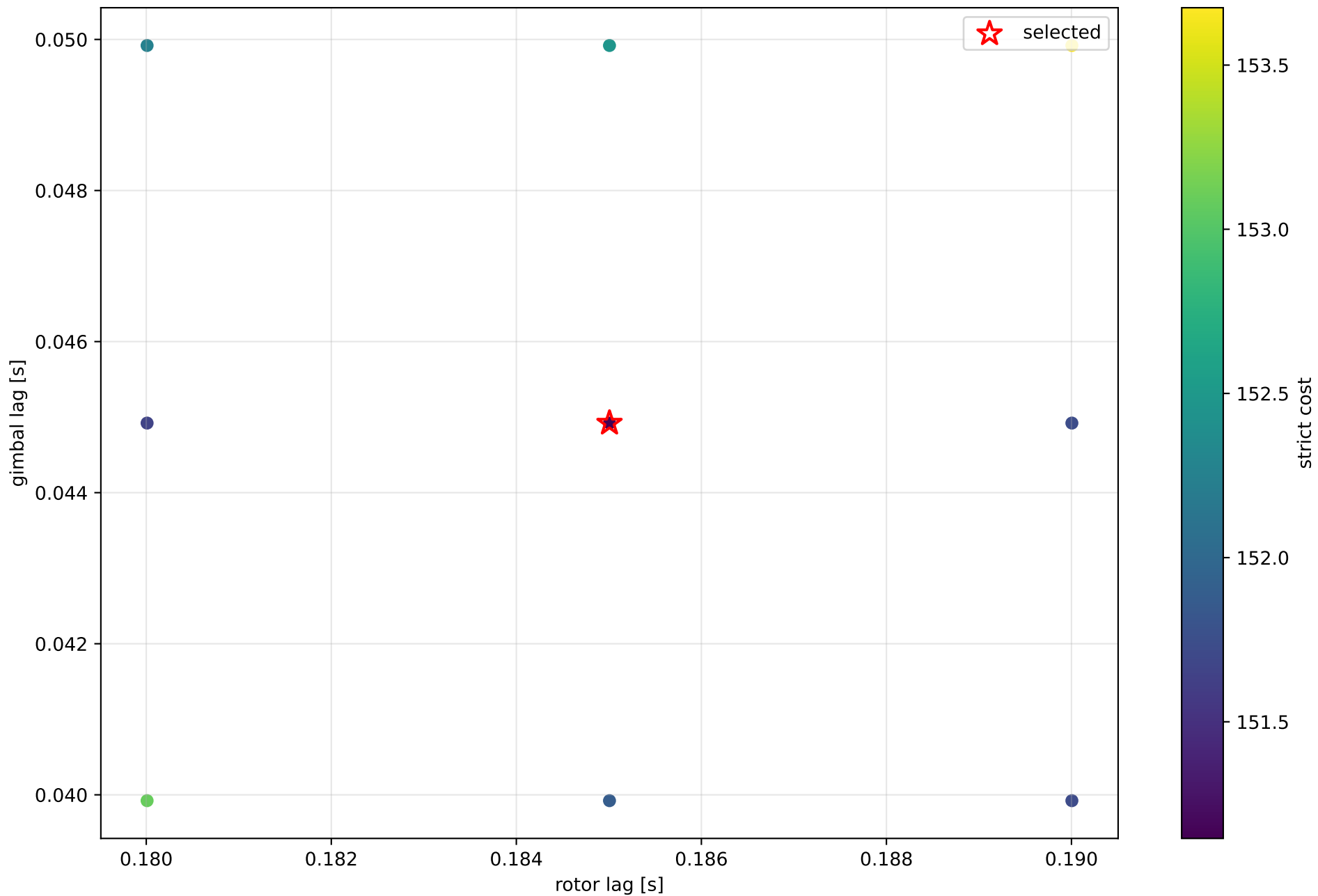


sweep_sg_window_covariance_005: parameter ridge



exact scale gauge: ||j v_scale||=1.02148e-13, rank=13, nullity=1

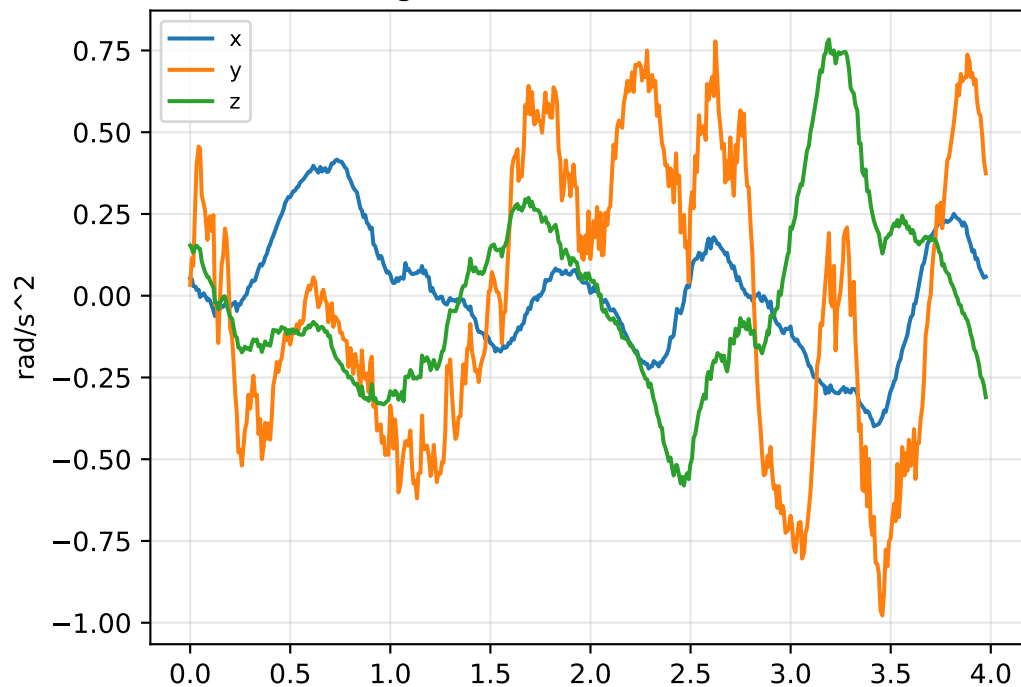
sweep_sg_window_covariance__005: strict lag candidates



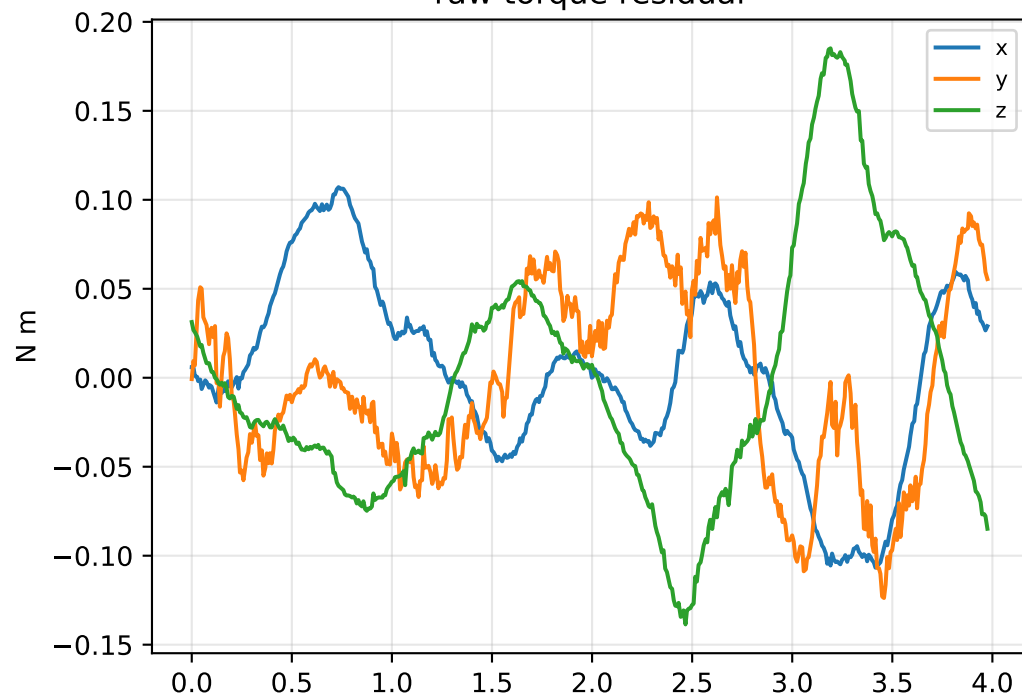
boundary flags: rotor lower=False upper=False; gimbal lower=False upper=False

sweep_sg_window_covariance_005: acceleration/wrench closure

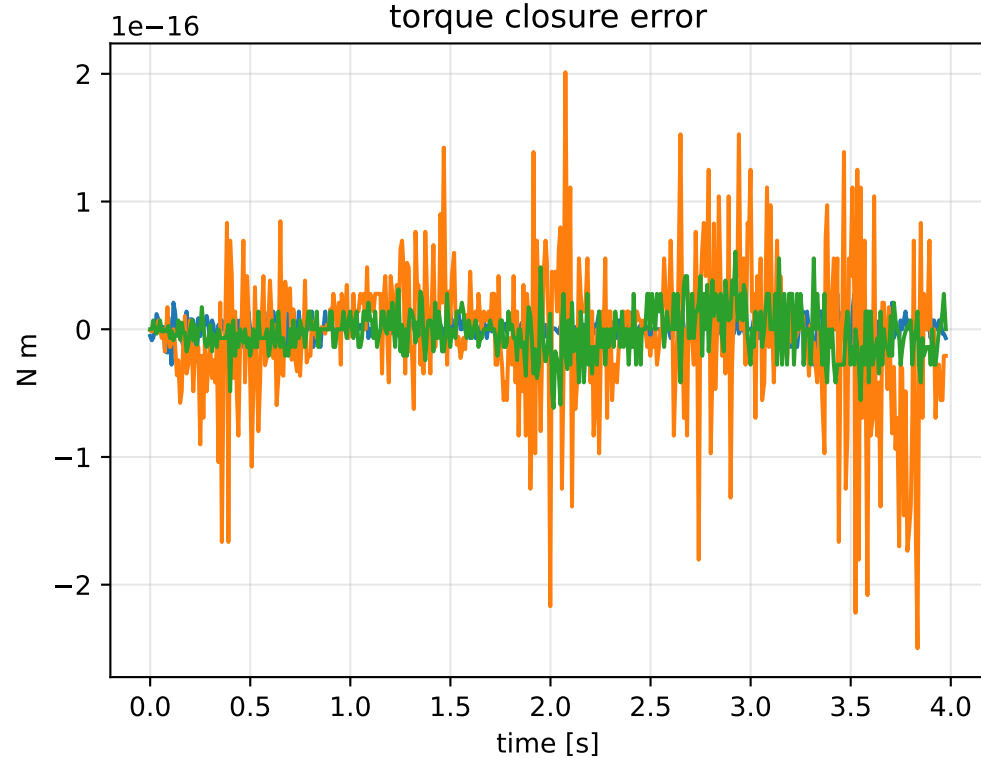
angular-acceleration residual



raw torque residual



torque closure error



principal moments [kg m²]
 nominal: [0.064953 0.065 0.128992]
 estimated: [0.111692 0.1916 0.281032]
 estimated / nominal: [1.719595 2.947694 2.178676]

force closure max/rms: 3.33067e-15 / 6.61894e-16
 torque closure max/rms: 2.498e-16 / 3.47394e-17